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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001

AMBM401 - MATHS FOR COMPETITVE EXAM

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define symmetric difference of the two sets A and B.
2. Find the equations of the line from the points (2, 2) and (3, 3).
3. Define Elasticity of Demand.
4. If $A = \begin{pmatrix} 4 & -2 \\ 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$ find AB.
5. What is meant by the true discount and Bankers discount?

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. State and prove Demorgan's law using Venn diagram.
7. A straight line passes through the point (-4, 9) and is such that the portion of it intercepted between the axes divided at the point in the ratio 3: 2. Find the equation.
8. Evaluate $\lim_{x\to\infty} \frac{2x^2+3x+5}{-5x^2+7x+9}$.
9. Find the Adjoint of the matrix $A = \begin{pmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{pmatrix}$.

10. Find present value of Rs. 2000, due in 8 months at 12% simple interest.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) In a survey of 100 students , the number studying various languages were found to be , English 28, German 30, French 42, English and German 8, English and French 10, German and French 5, and all the three languages 3.
i) How many students did not study any language?
ii) How many students had French as their only language?

(or)

b) Write down the properties of relation and also explain with suitable examples.

12. a) Find the equilibrium price and quantity for the following supply curve : $y = x^2 + 5x + 1$ and demand curve : y = 9- $2x^2$.

(or)

b) A company estimates that when its sales is Rs 60,000 ,its variable expense will be Rs 30,000 for a fixed expense of Rs 10,000.Find the break even point . What is the profit when the sales is Rs 50,000.

13. a) The total cost C, of making x units of a product is C= $0.00003x^3$- $0.045x^2$ +8x+25,000. Find the marginal cost at 1000 units out- put.

(or)

b) A television manufacturer produces x units in a production run at a total cost of Manufacturing and stocking given by

C= $25x + \frac{10^6}{x}$

a) Find $\frac{dc}{dx}$

b) Find the value of x at which $\frac{dc}{dx} = 0$

14. a) If $A=\begin{pmatrix} 2 & 5 & -1 \\ 3 & -1 & 2 \\ 7 & 2 & -3 \end{pmatrix}$ find A^{-1}.

(or)

b) Solve $x+y+z = 4$
$2x-y+z=3$, $x-2y+3z = 5$ by cramer's rule.

15. a) Two firms A and B have the same list price for an item..A offers discount of 25 % and 15% and B offers discount of 20%, 10% and 10%. Which is the more advantageous to the purchaser?

(or)

b) A certain amount of money was invested at 8% simple interest and after 9 month an equal amount was invested at 10% simple interest. Find the period in which the amount in each case becomes Rs 2,600.How much money was invested in each case.

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